Emory Peck

Engineering & Design Portfolio

Mechanical Engineering & Anthropology

Architecture Minor · Vanderbilt University

I am a junior at Vanderbilt University studying Mechanical Engineering and Anthropology, with a minor in Architecture. I enjoy projects that let me combine technical problem-solving with creativity, from mechanical prototypes and fabrication to architectural design.

I'm particularly interested in how the things we build affect the people who use them.



TECHNICAL SKILLS

Design	SolidWorks, CAD Modeling, Hand Drafting
Fabrication	3D Printing, Carbon Fiber Layup, Welding
Electronics	Arduino, Circuit Assembly, Prototyping
Analysis	Statics, Structural Analysis, Mechanical Testing

CONTACT INFORMATION

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Wearable Walking Robot Prototype

Assistive Technology / CAD / Digital Fabrication
Harvard PHYS S-12 |2023 | Independent Project



Project Overview

Independently designed and fabricated a robotic walking mechanism to explore assistive mobility. Developed custom mechanical linkages using Fusion 360 and 3D printing, integrating four servo motors into a dual-leg prototype.

Design Challenges

An initial stepper-motor design achieved only approximately 30° of rotation, below the intended 180° range. After experimenting with a gearbox, I redesigned the mechanism using servo motors to control the hip and knee joints.



Figure 1. Concept Sketches

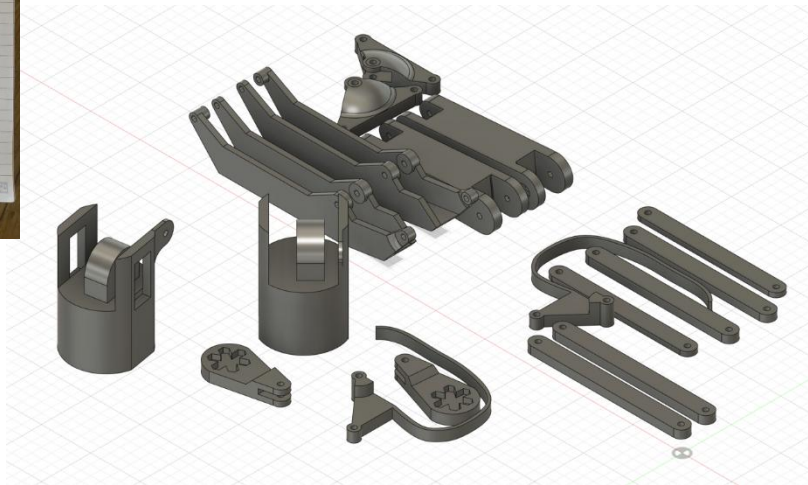


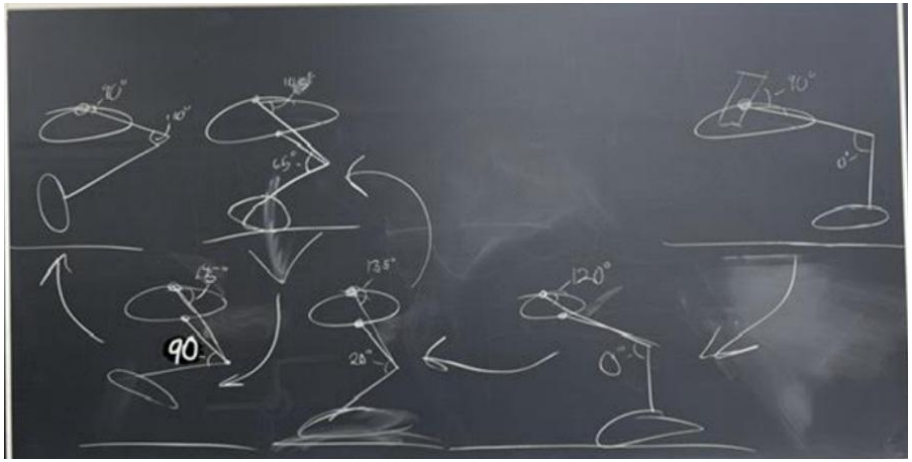
Figure 2. Fusion 360 Mechanical Components



Figure 3. Fabricated Linkage

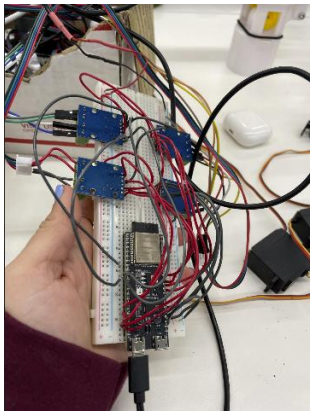
GAIT ANALYSIS

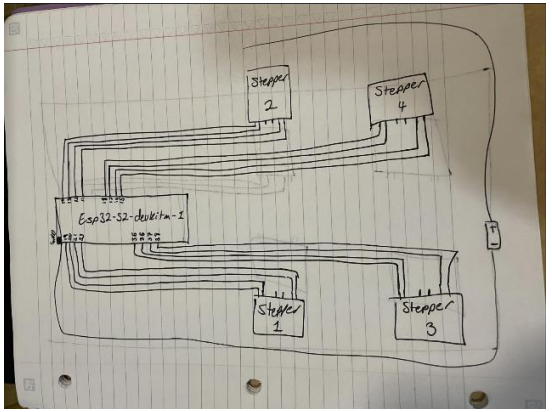
Studied photographs and video of the human gait to identify four phases of movement. Used measured hip and knee angles to establish target mechanical positions.



CONTROL SYSTEM

Programmed four servo motors using Arduino. Adjusted motor angles to account for opposite mounting orientations and introduced staggered movement to produce alternating leg cycles.





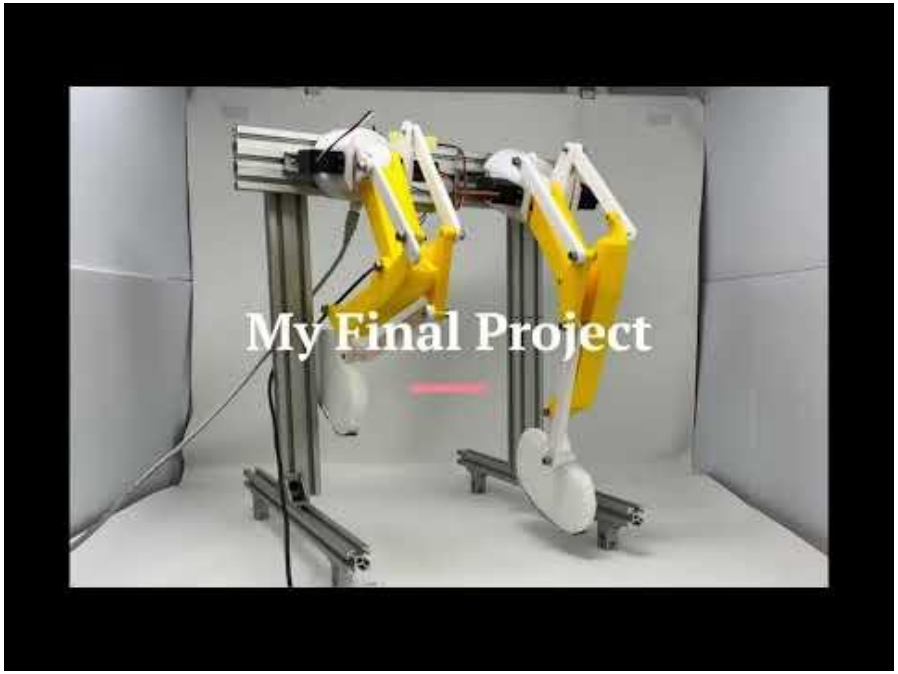
PROTOTYPE OUTCOME

Completed and tested a physical dual-leg robotic mechanism with programmed alternating movement. The project demonstrated integration of mechanical design, additive manufacturing, electronics, and motion control.

Key Takeaways

- Learned to troubleshoot mechanical issues and redesign parts when needed.
- Used gait observations to program realistic leg movements.
- Combined 3D printing, electronics, and coding to build and test a working prototype.

Final Prototype Demonstration



Vanderbilt Rocket Team

Carbon Fiber Manufacturing / CAD / Composite Materials | 2026

PROJECT OVERVIEW

Worked with Vanderbilt Rocket Team on manufacturing carbon fiber tubes for the rocket's airframe. I also researched epoxy materials and testing methods to help evaluate the strength of bonded components.

MANUFACTURING PROCESS

Participated in carbon fiber layup, epoxy application, curing, and finishing. Worked with teammates to apply carbon fiber around cylindrical molds and prepare the parts for further assembly.



Figure 4. Carbon Fiber Manufacturing Setup



Figure 5. Carbon Fiber Layup



Figure 6. Carbon Fiber Lower Rocket Body

Epoxy Testing Research
Researched different epoxy options, their strength, and how much weight bonded components could support. I explored application and curing methods to help determine how the epoxy should be used. Another team member carried out the physical testing.

Project Outcome
Contributed to the fabrication of carbon fiber tubes and the research supporting the team's epoxy testing. Gained experience with composite manufacturing techniques and working on a larger engineering project.

Key Takeaways

- Learned how to prepare, lay-up, and finish carbon fiber tubes.
- Gained a better understanding of how epoxy affects composite construction.
- Worked with teammates throughout the manufacturing process.

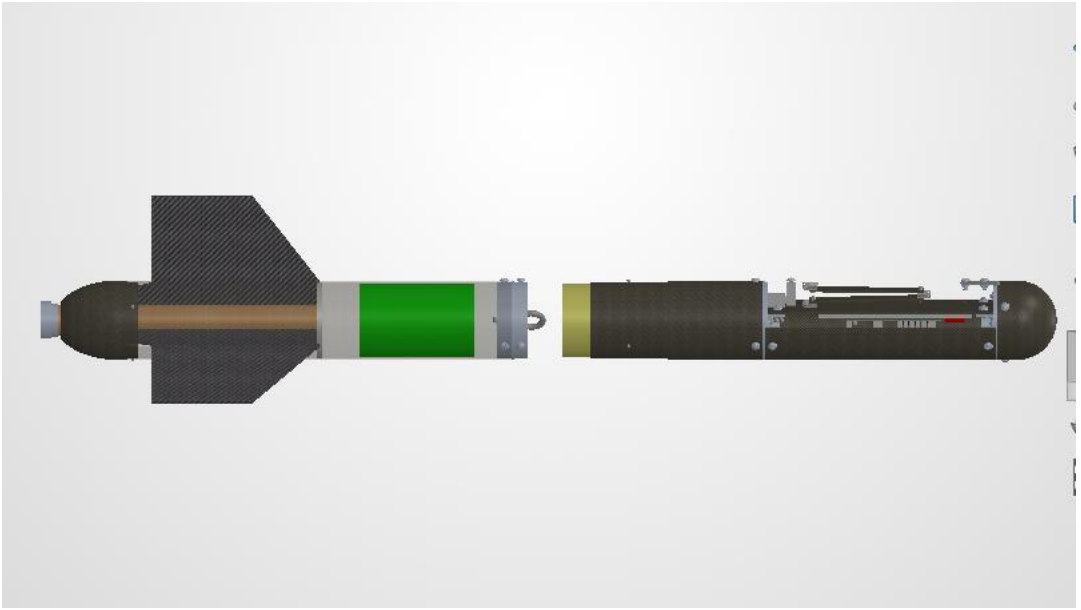


Figure 7. Rocket CAD Assembly



Figure 8. Epoxy Bond Load Testing (33 lb)

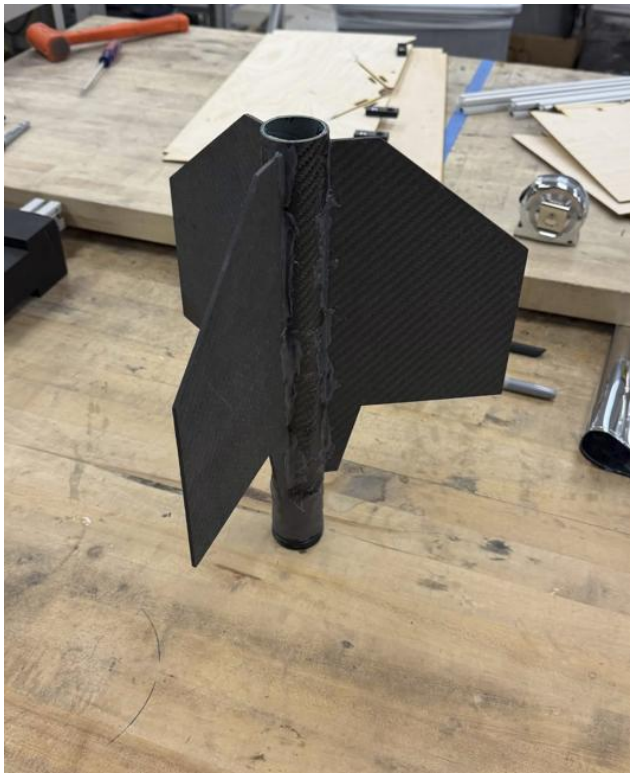


Figure 9. Composite Fin Assembly

Mexico City Public Library

Architectural Design / Site Analysis / Hand Drafting

Oxford Summer Courses — Architecture | 2022
Individual Conceptual Design Project

PROJECT OVERVIEW

For my final architecture project at Oxford Summer Courses, I was given a site in Mexico City and asked to design a building that would benefit the surrounding community. I chose a public library because I wanted to create a welcoming place where people could read, study, and spend time.

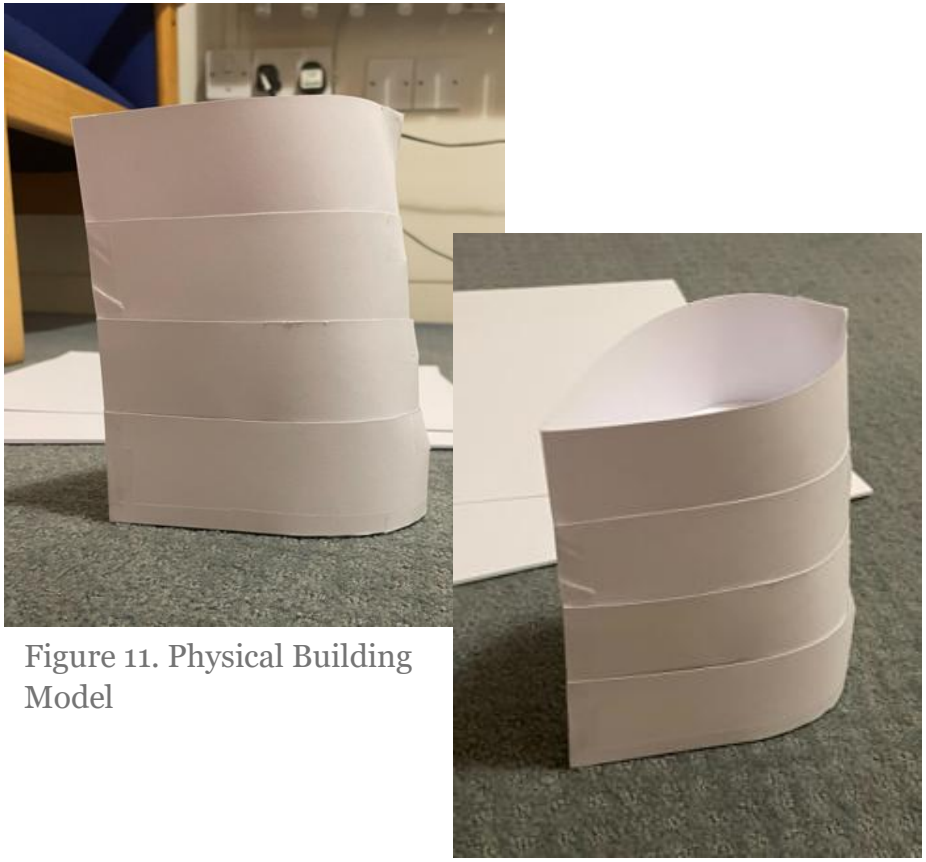


Figure 11. Physical Building Model

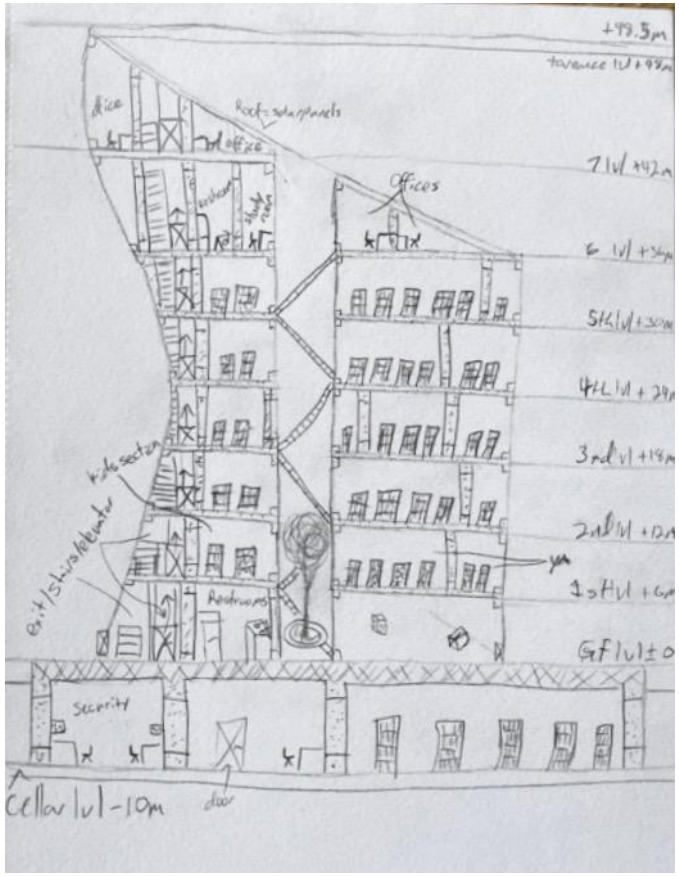


Figure 10. Library Section Drawing

INSPIRATION

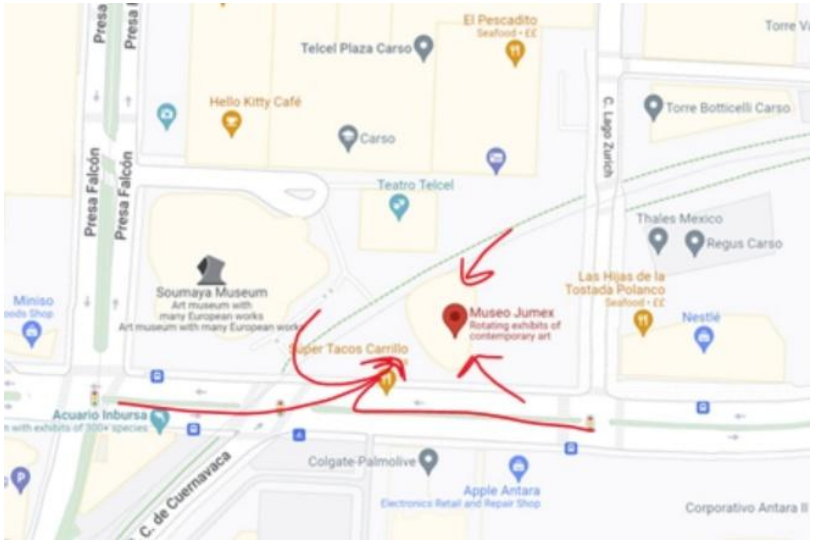
I've always loved unusual, nature-inspired buildings, especially Singapore's Supertree Grove and Gaudí's architecture. I wanted my library to feel almost like a living structure, with curved walls, greenery, and open spaces that would make people want to explore it.

DESIGN CONCEPT

Designed a nine-floor library with a curved exterior, central spiral staircase, study spaces, and reading areas. I also explored natural lighting, solar panels, and landscaping throughout the building.

SITE ANALYSIS

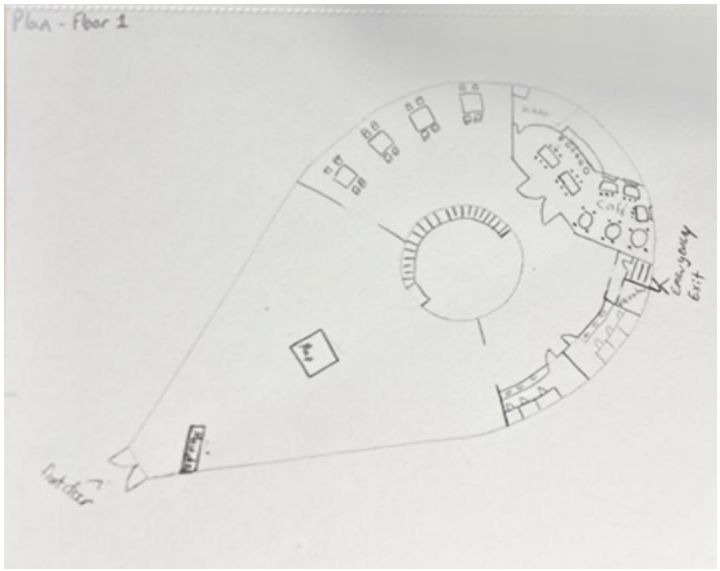
Studied the surrounding museums, apartments, businesses, and pedestrian access to understand how the building would fit into the neighborhood.

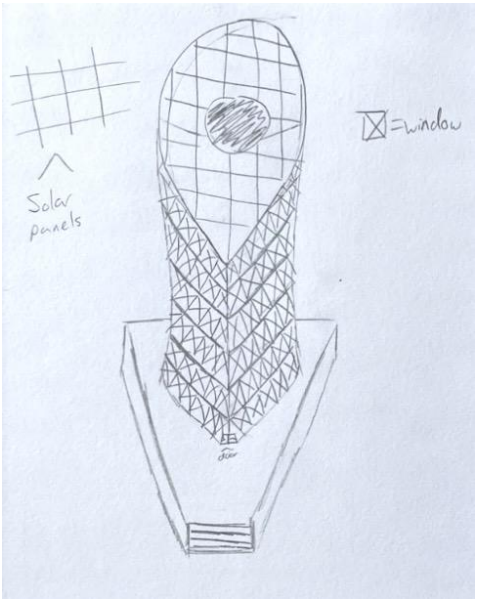


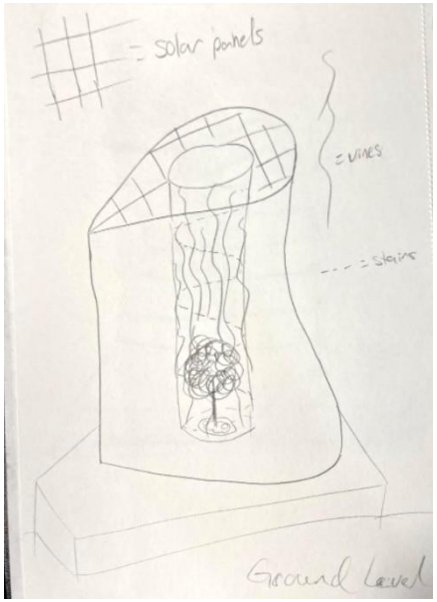


BUILDING DESIGN AND SUSTAINABILITY

I incorporated ideas such as solar panels, natural lighting, planted walkways, and vegetation along the central staircase. My goal was to make nature a part of the building rather than simply adding landscaping around it.







Steel Truss Design & Testing

Structural Analysis / Welding / Load Testing

Vanderbilt University — CE2200 Statics | Fall 2025 | Team Project

PROJECT OVERVIEW

For our CE2200 Statics final project, we designed, welded, and tested a steel truss representing a proposed highway cap over I-40 in Nashville. We calculated internal forces, evaluated potential failure modes, and compared our predictions with physical testing.

COMMUNITY CONTEXT

Alongside the structural work, we researched how the construction of I-40 divided Nashville's Jefferson Street community. Our proposed highway cap included green space, a community building, and a playground to explore how new infrastructure could help reconnect the neighborhood.



Figure 12. Welded Steel Truss Connection



Figure 11. Team Photo with Completed Truss

DESIGN & ANALYSIS

Used the method of joints and method of sections to calculate internal forces under a 350 lb design load. We checked our calculations using the Johns Hopkins Truss Calculator and estimated the buckling and yielding limits of the steel members.

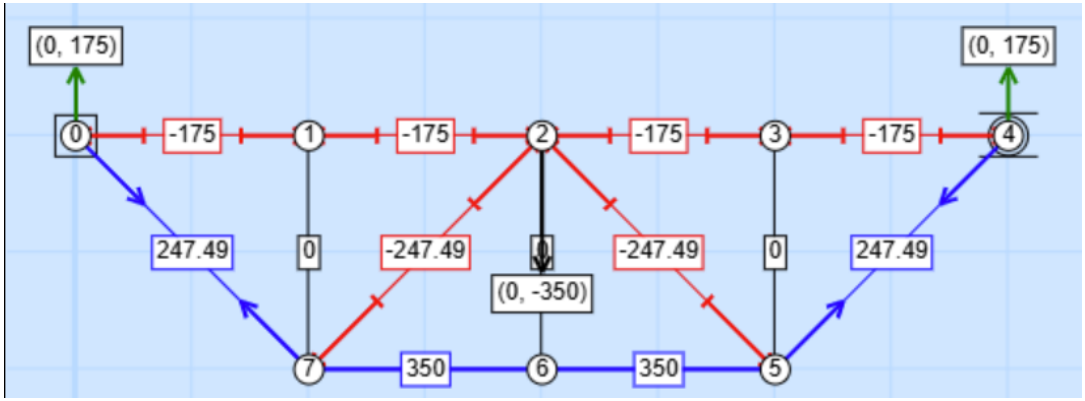


Figure 13. John Hopkins Simulation, Calculated Internal Truss Forces

FABRICATION & TESTING

Assembled the truss using pre-cut steel members and connector plates, tack-welded the connections, and reinforced welds where needed. During testing, the first visible buckling occurred around 150 lb, and the truss eventually supported approximately 400 lb before significant deformation.



Figure 14. Completed Steel Truss

RESULTS

The experiment showed that buckling was the main failure mode, as predicted by our calculations. However, the physical truss began buckling earlier than expected, likely due to differences between the ideal model and the welded structure.

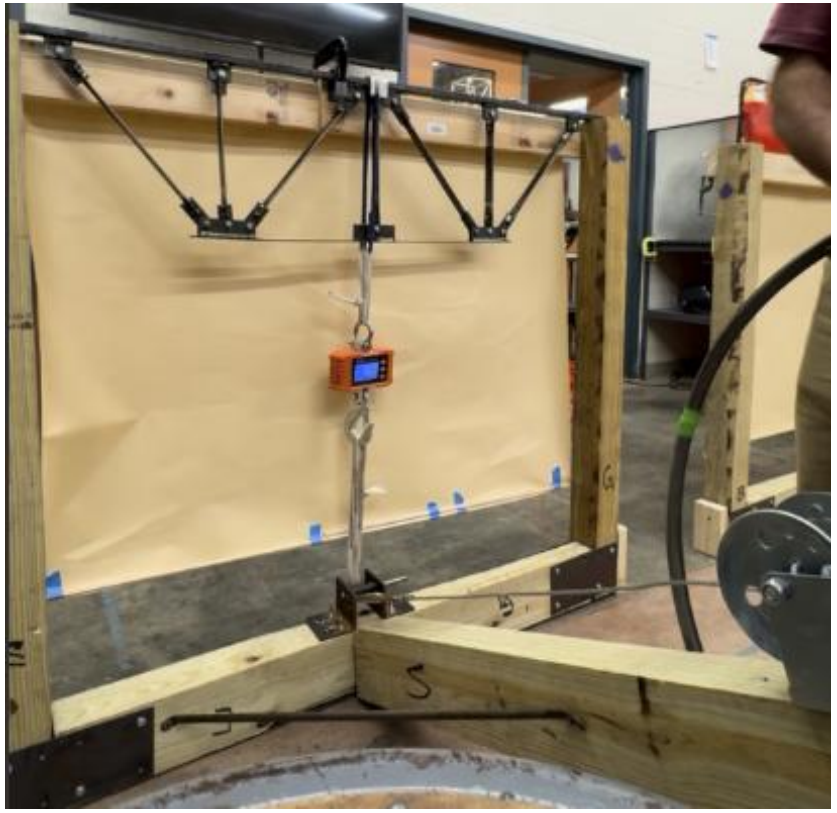


Figure 15. Truss Load Testing Setup

KEY TAKEAWAYS

- Applied statics calculations to a physical steel structure.
- Gained experience with welding, assembly, and load testing.
- Compared predicted failure loads with actual results.